```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

```
df = pd.read_csv('/content/laptop_data .csv')
```

```
df.head()
```

	Unnamed: 0	Company	TypeName	Inches	ScreenResolution	Cpu	Ram	Memory	Gpu	OpSys	Weight	Price
0	0	Apple	Ultrabook	13.3	IPS Panel Retina Display 2560x1600	Intel Core i5 2.3GHz	8GB	128GB SSD	Intel Iris Plus Graphics 640	macOS	1.37kg	71378.6832
1	1	Apple	Ultrabook	13.3	1440x900	Intel Core i5 1.8GHz	8GB	128GB Flash Storage	Intel HD Graphics 6000	macOS	1.34kg	47895.5232
						Intel Core i5						

Next steps: Generate code with df | View recommended plots | New interactive sheet

```
df.shape
```

```
(1303, 12)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1303 entries, 0 to 1302
Data columns (total 12 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   Unnamed: 0        1303 non-null   int64
 1   Company           1303 non-null   object
 2   TypeName          1303 non-null   object
 3   Inches            1303 non-null   float64
 4   ScreenResolution  1303 non-null   object
 5   Cpu               1303 non-null   object
 6   Ram               1303 non-null   object
 7   Memory            1303 non-null   object
 8   Gpu               1303 non-null   object
 9   OpSys             1303 non-null   object
 10  Weight            1303 non-null   object
 11  Price             1303 non-null   float64
dtypes: float64(2), int64(1), object(9)
memory usage: 122.3+ KB
```

```
df.duplicated().sum()
```

```
0
```

```
df.isnull().sum()
```

	0
Unnamed: 0	0
Company	0
TypeName	0
Inches	0
ScreenResolution	0
Cpu	0
Ram	0
Memory	0
Gpu	0
OpSys	0
Weight	0
Price	0

dtype: int64

```
df.drop(columns=['Unnamed: 0'],inplace=True)
```

```
df.head()
```

	Company	TypeName	Inches	ScreenResolution	Cpu	Ram	Memory	Gpu	OpSys	Weight	Price
0	Apple	Ultrabook	13.3	IPS Panel Retina Display 2560x1600	Intel Core i5 2.3GHz	8GB	128GB SSD	Intel Iris Plus Graphics 640	macOS	1.37kg	71378.6832
1	Apple	Ultrabook	13.3	1440x900	Intel Core i5 1.8GHz	8GB	128GB Flash Storage	Intel HD Graphics 6000	macOS	1.34kg	47895.5232
2	HP	Notebook	15.6	Full HD 1920x1080	Intel Core i5 7200U 2.5GHz	8GB	256GB SSD	Intel HD Graphics 620	No OS	1.86kg	30636.0000

Next steps: Generate code with df | View recommended plots | New interactive sheet

```
df['Ram'] = df['Ram'].str.replace('GB','')
df['Weight'] = df['Weight'].str.replace('kg','')
```

```
df.head()
```

	Company	TypeName	Inches	ScreenResolution	Cpu	Ram	Memory	Gpu	OpSys	Weight	Price
0	Apple	Ultrabook	13.3	IPS Panel Retina Display 2560x1600	Intel Core i5 2.3GHz	8	128GB SSD	Intel Iris Plus Graphics 640	macOS	1.37	71378.6832
1	Apple	Ultrabook	13.3	1440x900	Intel Core i5 1.8GHz	8	128GB Flash Storage	Intel HD Graphics 6000	macOS	1.34	47895.5232
2	HP	Notebook	15.6	Full HD 1920x1080	Intel Core i5 7200U 2.5GHz	8	256GB SSD	Intel HD Graphics 620	No OS	1.86	30636.0000

Next steps: Generate code with df | View recommended plots | New interactive sheet

```
df['Ram'] = df['Ram'].astype('int32')
df['Weight'] = df['Weight'].astype('float32')
```

```
df.info()
```

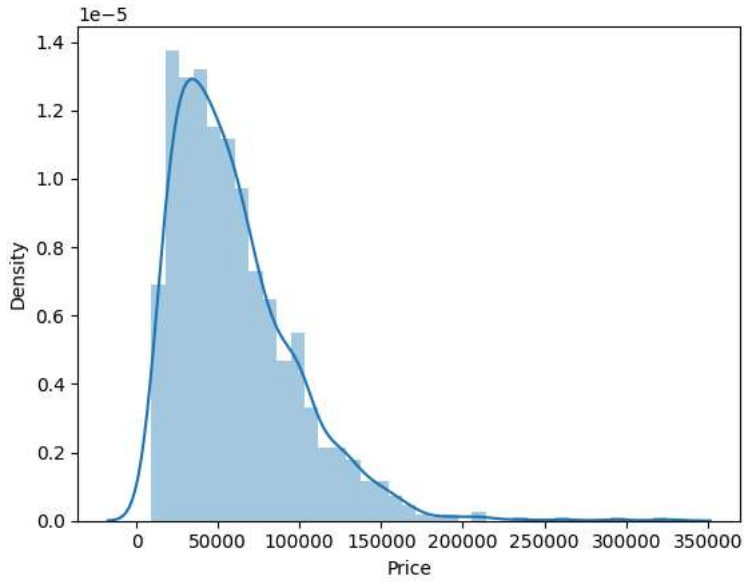
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1303 entries, 0 to 1302
Data columns (total 11 columns):
 #   Column            Non-Null Count  Dtype  
---  ------            --------------  -----  
 0   Company           1303 non-null   object 
 1   TypeName          1303 non-null   object 
 2   Inches            1303 non-null   float64
```

```
 3   ScreenResolution  1303 non-null   object
 4   Cpu               1303 non-null   object
 5   Ram               1303 non-null   int32
 6   Memory            1303 non-null   object
 7   Gpu               1303 non-null   object
 8   OpSys             1303 non-null   object
 9   Weight            1303 non-null   float32
 10  Price             1303 non-null   float64
dtypes: float32(1), float64(2), int32(1), object(7)
memory usage: 101.9+ KB
```

```
import seaborn as sns
```

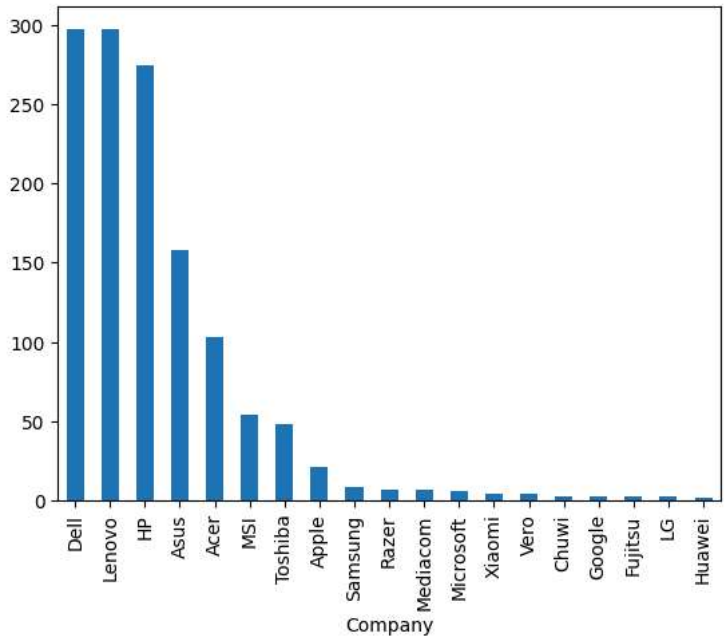
```
sns.distplot(df['Price'])
```

```
/tmp/ipython-input-834922981.py:1: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with
similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

  sns.distplot(df['Price'])
<Axes: xlabel='Price', ylabel='Density'>
```

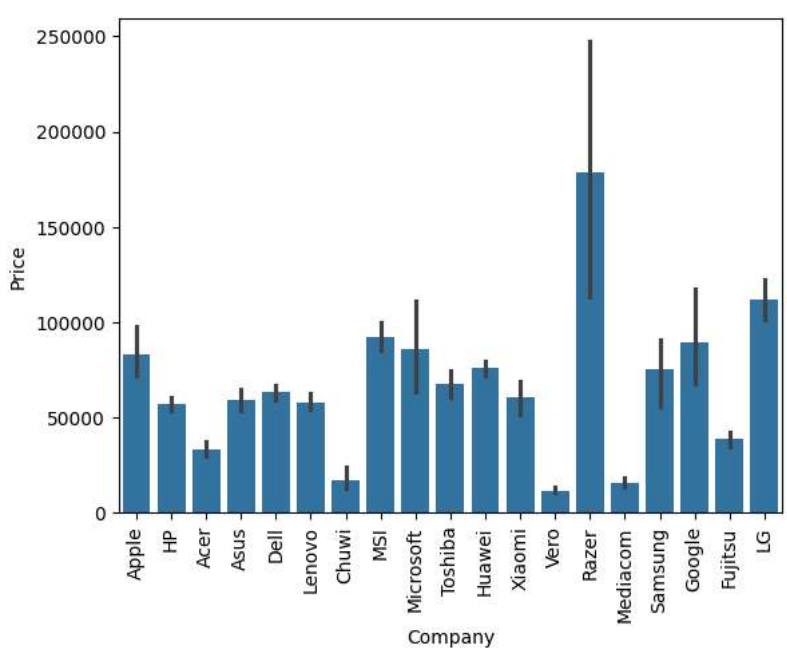



```
df['Company'].value_counts().plot(kind='bar')
```

```
<Axes: xlabel='Company'>
```

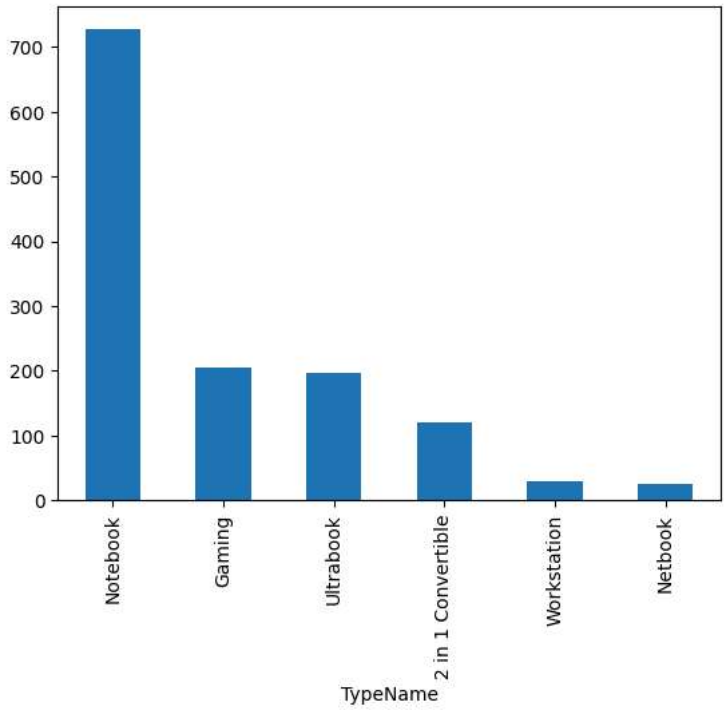



```
sns.barplot(x=df['Company'],y=df['Price'])
plt.xticks(rotation='vertical')
plt.show()
```

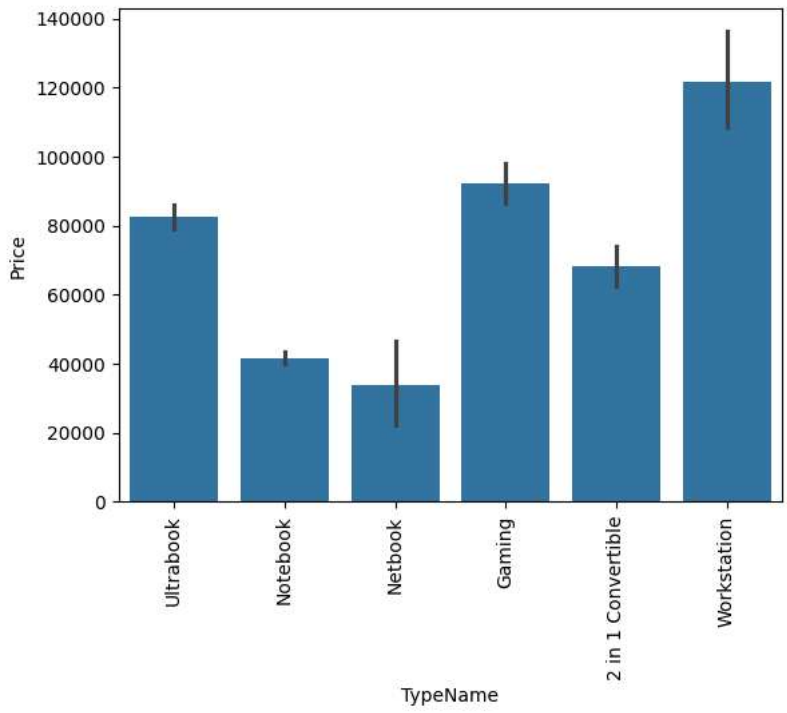



```
df['TypeName'].value_counts().plot(kind='bar')
```

```
<Axes: xlabel='TypeName'>
```

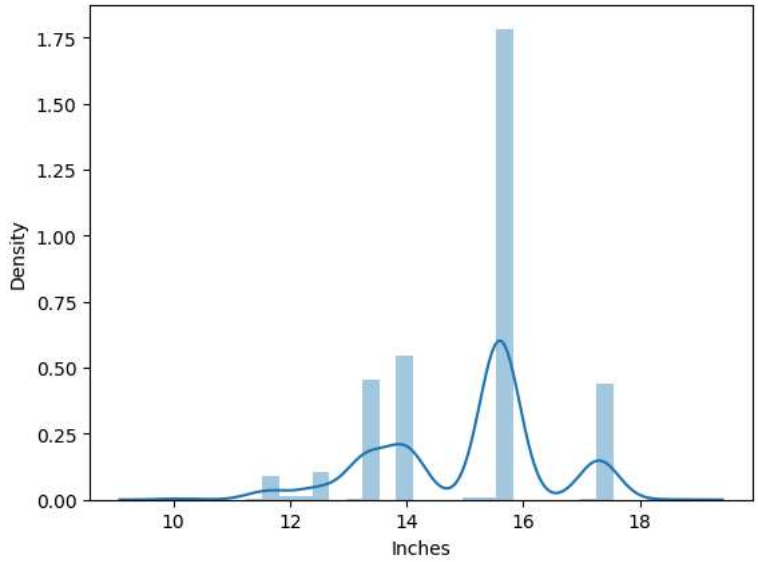



```
sns.barplot(x=df['TypeName'],y=df['Price'])
plt.xticks(rotation='vertical')
plt.show()
```



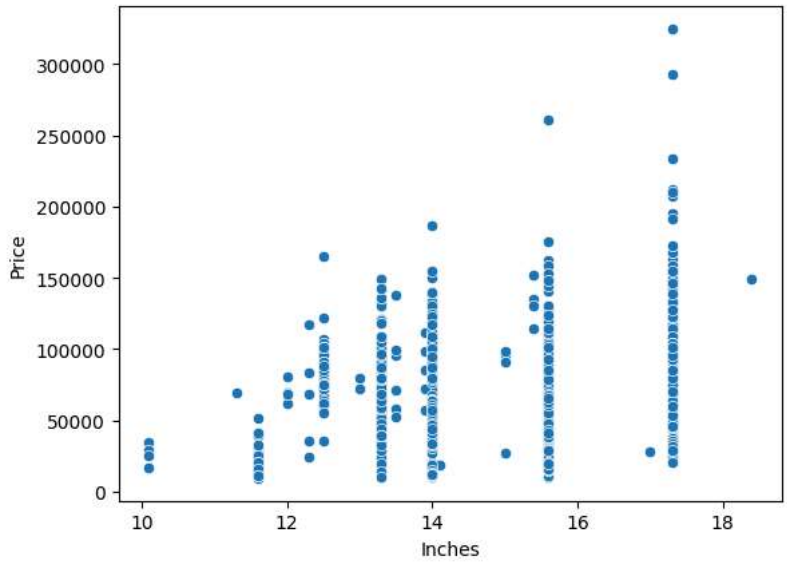

```
sns.distplot(df['Inches'])
```

```
/tmp/ipython-input-1439577752.py:1: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with
similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

  sns.distplot(df['Inches'])
<Axes: xlabel='Inches', ylabel='Density'>
```




```
sns.scatterplot(x=df['Inches'],y=df['Price'])
```

```
<Axes: xlabel='Inches', ylabel='Price'>
```




```
df['ScreenResolution'].value_counts()
```

ScreenResolution	count
Full HD 1920x1080	507
1366x768	281
IPS Panel Full HD 1920x1080	230
IPS Panel Full HD / Touchscreen 1920x1080	53
Full HD / Touchscreen 1920x1080	47
1600x900	23
Touchscreen 1366x768	16
Quad HD+ / Touchscreen 3200x1800	15
IPS Panel 4K Ultra HD 3840x2160	12
IPS Panel 4K Ultra HD / Touchscreen 3840x2160	11
4K Ultra HD / Touchscreen 3840x2160	10
4K Ultra HD 3840x2160	7
Touchscreen 2560x1440	7
IPS Panel 1366x768	7
IPS Panel Quad HD+ / Touchscreen 3200x1800	6
IPS Panel Retina Display 2560x1600	6
IPS Panel Retina Display 2304x1440	6
Touchscreen 2256x1504	6
IPS Panel Touchscreen 2560x1440	5
IPS Panel Retina Display 2880x1800	4
IPS Panel Touchscreen 1920x1200	4
1440x900	4
IPS Panel 2560x1440	4
IPS Panel Quad HD+ 2560x1440	3
Quad HD+ 3200x1800	3
1920x1080	3
Touchscreen 2400x1600	3
2560x1440	3
IPS Panel Touchscreen 1366x768	3
IPS Panel Touchscreen / 4K Ultra HD 3840x2160	2
IPS Panel Full HD 2160x1440	2
IPS Panel Quad HD+ 3200x1800	2
IPS Panel Retina Display 2736x1824	1
IPS Panel Full HD 1920x1200	1
IPS Panel Full HD 2560x1440	1
IPS Panel Full HD 1366x768	1
Touchscreen / Full HD 1920x1080	1
Touchscreen / Quad HD+ 3200x1800	1
Touchscreen / 4K Ultra HD 3840x2160	1
IPS Panel Touchscreen 2400x1600	1

dtype: int64

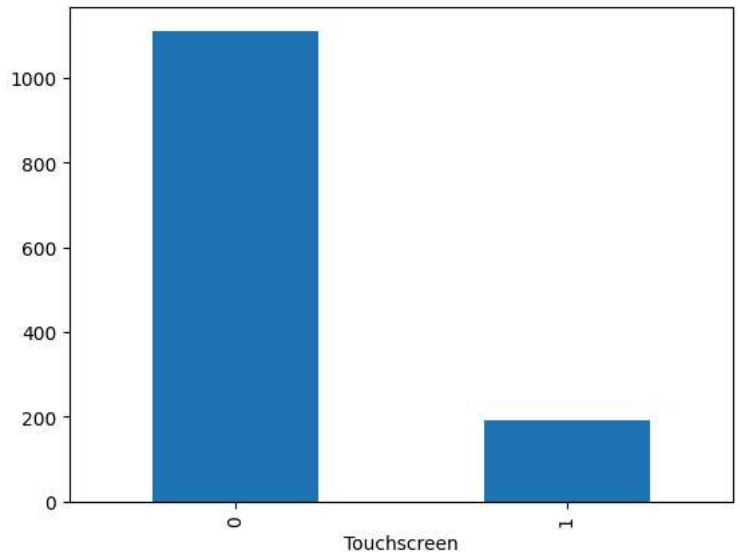
```
df['Touchscreen'] = df['ScreenResolution'].apply(lambda x:1 if 'Touchscreen' in x else 0)
```

```
df.sample(5)
```

	Company	TypeName	Inches	ScreenResolution	Cpu	Ram	Memory	Gpu	OpSys	Weight	Price	Touchscreen
570	HP	Workstation	17.3	1600x900	Intel Core i5 7440HQ 2.8GHz	8	500GB HDD	Nvidia Quadro M1200	Windows 10	3.14	99153.5472	0
486	Dell	Workstation	15.6	IPS Panel Full HD 1920x1080	Intel Core i7 6820HQ 2.7GHz	16	512GB SSD	Nvidia Quadro M620	Windows 10	2.17	124568.6400	0

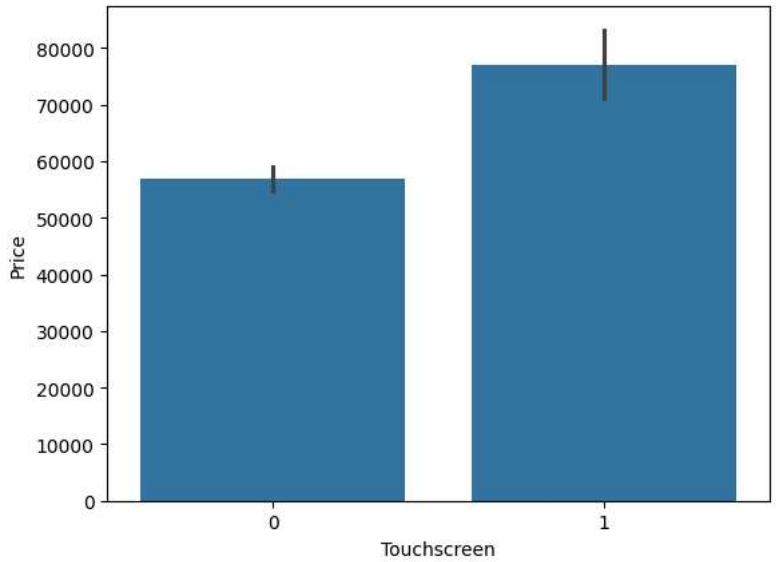
```
df['Touchscreen'].value_counts().plot(kind='bar')
```

```
<Axes: xlabel='Touchscreen'>
```




```
sns.barplot(x=df['Touchscreen'],y=df['Price'])
```

```
<Axes: xlabel='Touchscreen', ylabel='Price'>
```




```
df['Ips'] = df['ScreenResolution'].apply(lambda x:1 if 'IPS' in x else 0)
```

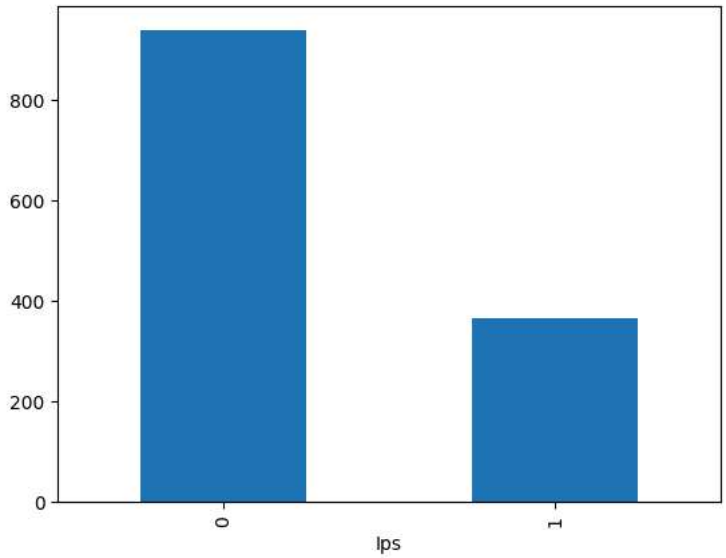
```
df.head()
```

	Company	TypeName	Inches	ScreenResolution	Cpu	Ram	Memory	Gpu	OpSys	Weight	Price	Touchscreen	Ips
0	Apple	Ultrabook	13.3	IPS Panel Retina Display 2560x1600	Intel Core i5 2.3GHz	8	128GB SSD	Intel Iris Plus Graphics 640	macOS	1.37	71378.6832	0	1
1	Apple	Ultrabook	13.3	1440x900	Intel Core i5 1.8GHz	8	128GB Flash Storage	Intel HD Graphics 6000	macOS	1.34	47895.5232	0	0

Next steps: Generate code with df | View recommended plots | New interactive sheet

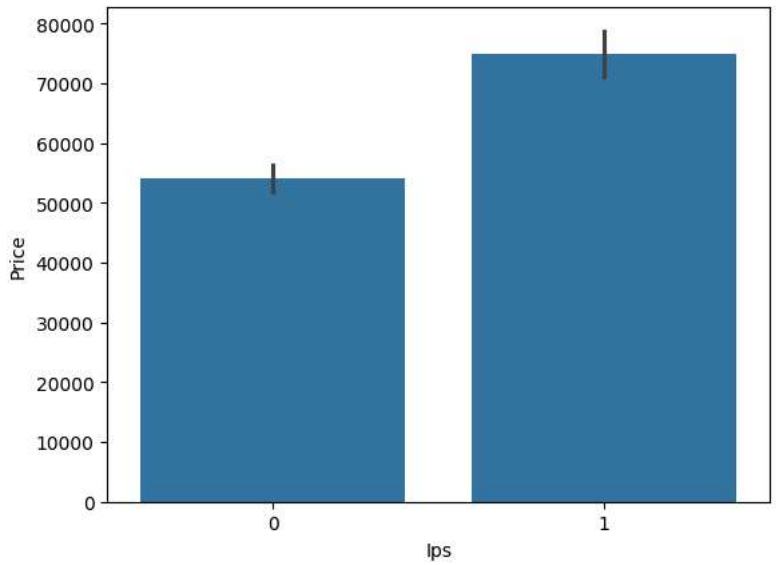
```
df['Ips'].value_counts().plot(kind='bar')
```

```
<Axes: xlabel='Ips'>
```




```
sns.barplot(x=df['Ips'],y=df['Price'])
```

```
<Axes: xlabel='Ips', ylabel='Price'>
```




```
new = df['ScreenResolution'].str.split('x',n=1,expand=True)


df['X_res'] = new[0]
df['Y_res'] = new[1]


df.sample(5)
```

	Company	TypeName	Inches	ScreenResolution	Cpu	Ram	Memory	Gpu	OpSys	Weight	Price	Touchscreen	Ips	X_res
1263	Acer	Notebook	15.6	1366x768	Intel Celeron Dual Core N3060 1.6GHz	4	500GB HDD	Intel HD Graphics 400	Linux	2.40	15397.9200	0	0	1366
					Intel Core i7		512GB	Intel HD	Windows					Full

```
df['X_res'] = df['X_res'].str.replace(',','').str.findall(r'(\d+\.?\d+)').apply(lambda x:x[0])
```

```
df.head()
```

	Company	TypeName	Inches	ScreenResolution	Cpu	Ram	Memory	Gpu	OpSys	Weight	Price	Touchscreen	Ips	X_res	Y_res
0	Apple	Ultrabook	13.3	IPS Panel Retina Display 2560x1600	Intel Core i5 2.3GHz	8	128GB SSD	Intel Iris Plus Graphics 640	macOS	1.37	71378.6832	0	1	2560	1600
1	Apple	Ultrabook	13.3	1440x900	Intel Core i5 1.8GHz	8	128GB Flash Storage	Intel HD Graphics 6000	macOS	1.34	47895.5232	0	0	1440	900

Next steps: Generate code with df | View recommended plots | New interactive sheet

```
df['X_res'] = df['X_res'].astype('int')
df['Y_res'] = df['Y_res'].astype('int')
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1303 entries, 0 to 1302
Data columns (total 15 columns):
 #   Column            Non-Null Count  Dtype  
---  ------            --------------  -----  
 0   Company           1303 non-null   object 
 1   TypeName          1303 non-null   object 
 2   Inches            1303 non-null   float64
 3   ScreenResolution  1303 non-null   object 
 4   Cpu               1303 non-null   object 
 5   Ram               1303 non-null   int32  
 6   Memory            1303 non-null   object 
 7   Gpu               1303 non-null   object 
 8   OpSys             1303 non-null   object 
 9   Weight            1303 non-null   float32
 10  Price             1303 non-null   float64
 11  Touchscreen       1303 non-null   int64  
 12  Ips               1303 non-null   int64  
 13  X_res             1303 non-null   int64  
 14  Y_res             1303 non-null   int64  
dtypes: float32(1), float64(2), int32(1), int64(4), object(7)
memory usage: 142.6+ KB
```

```
df.corr(numeric_only=True)['Price']
```

	Price
Inches	0.068197
Ram	0.743007
Weight	0.210370
Price	1.000000
Touchscreen	0.191226
Ips	0.252208
X_res	0.556529
Y_res	0.552809

dtype: float64

```
df['ppi'] = (((df['X_res']**2) + (df['Y_res']**2))**0.5/df['Inches']).astype('float')
```

```
df.corr(numeric_only=True)['Price']
```

	Price
Inches	0.068197
Ram	0.743007
Weight	0.210370
Price	1.000000
Touchscreen	0.191226
Ips	0.252208
X_res	0.556529
Y_res	0.552809
ppi	0.473487

dtype: float64

```
df.drop(columns=['ScreenResolution'],inplace=True)
```

```
df.head()
```

	Company	TypeName	Inches	Cpu	Ram	Memory	Gpu	OpSys	Weight	Price	Touchscreen	Ips	X_res	Y_res	ppi
0	Apple	Ultrabook	13.3	Intel Core i5 2.3GHz	8	128GB SSD	Intel Iris Plus Graphics 640	macOS	1.37	71378.6832	0	1	2560	1600	226.983005
1	Apple	Ultrabook	13.3	Intel Core i5 1.8GHz	8	128GB Flash Storage	Intel HD Graphics 6000	macOS	1.34	47895.5232	0	0	1440	900	127.677940

Next steps: Generate code with df | View recommended plots | New interactive sheet

```
df.drop(columns=['Inches','X_res','Y_res'],inplace=True)
```

```
df.head()
```

	Company	TypeName	Cpu	Ram	Memory	Gpu	OpSys	Weight	Price	Touchscreen	Ips	ppi
0	Apple	Ultrabook	Intel Core i5 2.3GHz	8	128GB SSD	Intel Iris Plus Graphics 640	macOS	1.37	71378.6832	0	1	226.983005
1	Apple	Ultrabook	Intel Core i5 1.8GHz	8	128GB Flash Storage	Intel HD Graphics 6000	macOS	1.34	47895.5232	0	0	127.677940
2	HP	Notebook	Intel Core i5 7200U 2.5GHz	8	256GB SSD	Intel HD Graphics 620	No OS	1.86	30636.0000	0	0	141.211998

Next steps: Generate code with df | View recommended plots | New interactive sheet

```
df['Cpu'].value_counts()
```

Cpu	count
Intel Core i5 7200U 2.5GHz	190
Intel Core i7 7700HQ 2.8GHz	146
Intel Core i7 7500U 2.7GHz	134
Intel Core i7 8550U 1.8GHz	73
Intel Core i5 8250U 1.6GHz	72
...	...
Intel Core M M3-6Y30 0.9GHz	1
AMD A9-Series 9420 2.9GHz	1
Intel Core i3 6006U 2.2GHz	1
AMD A6-Series 7310 2GHz	1
Intel Xeon E3-1535M v6 3.1GHz	1

118 rows × 1 columns

dtype: int64

```
df['Cpu Name'] = df['Cpu'].apply(lambda x:" ".join(x.split()[0:3]))
```

```
df.head()
```

	Company	TypeName	Cpu	Ram	Memory	Gpu	OpSys	Weight	Price	Touchscreen	Ips	ppi	Cpu Name
0	Apple	Ultrabook	Intel Core i5 2.3GHz	8	128GB SSD	Intel Iris Plus Graphics 640	macOS	1.37	71378.6832	0	1	226.983005	Intel Core i5
1	Apple	Ultrabook	Intel Core i5 1.8GHz	8	128GB Flash Storage	Intel HD Graphics 6000	macOS	1.34	47895.5232	0	0	127.677940	Intel Core i5
			Intel Core i5										

Next steps: Generate code with df | View recommended plots | New interactive sheet

```
def fetch_processor(text):
    if text == 'Intel Core i7' or text == 'Intel Core i5' or text == 'Intel Core i3':
        return text
    else:
        if text.split()[0] == 'Intel':
            return 'Other Intel Processor'
        else:
            return 'AMD Processor'
```

```
df['Cpu brand'] = df['Cpu Name'].apply(fetch_processor)
```

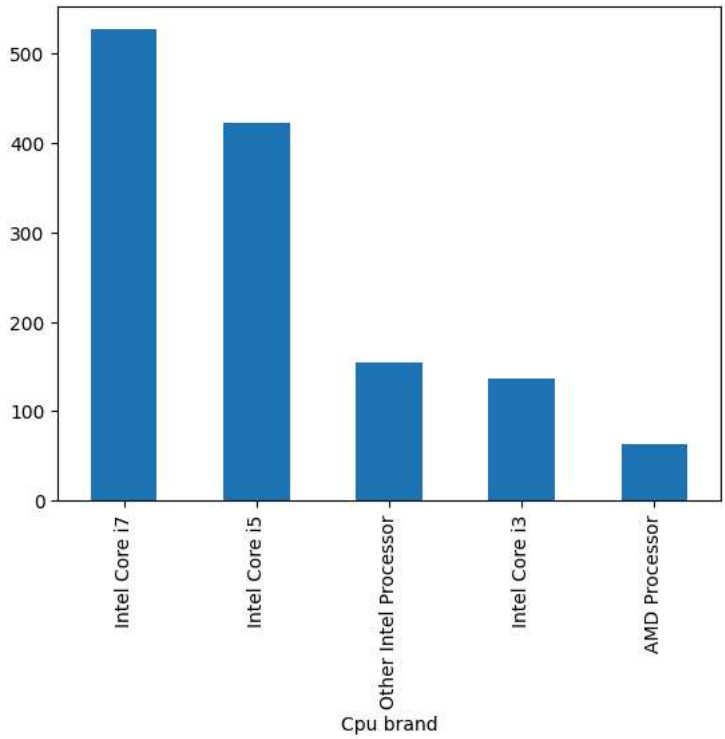
```
df.head()
```

	Company	TypeName	Cpu	Ram	Memory	Gpu	OpSys	Weight	Price	Touchscreen	Ips	ppi	Cpu Name	Cpu brand
0	Apple	Ultrabook	Intel Core i5 2.3GHz	8	128GB SSD	Intel Iris Plus Graphics 640	macOS	1.37	71378.6832	0	1	226.983005	Intel Core i5	Intel Core i5
1	Apple	Ultrabook	Intel Core i5 1.8GHz	8	128GB Flash Storage	Intel HD Graphics 6000	macOS	1.34	47895.5232	0	0	127.677940	Intel Core i5	Intel Core i5

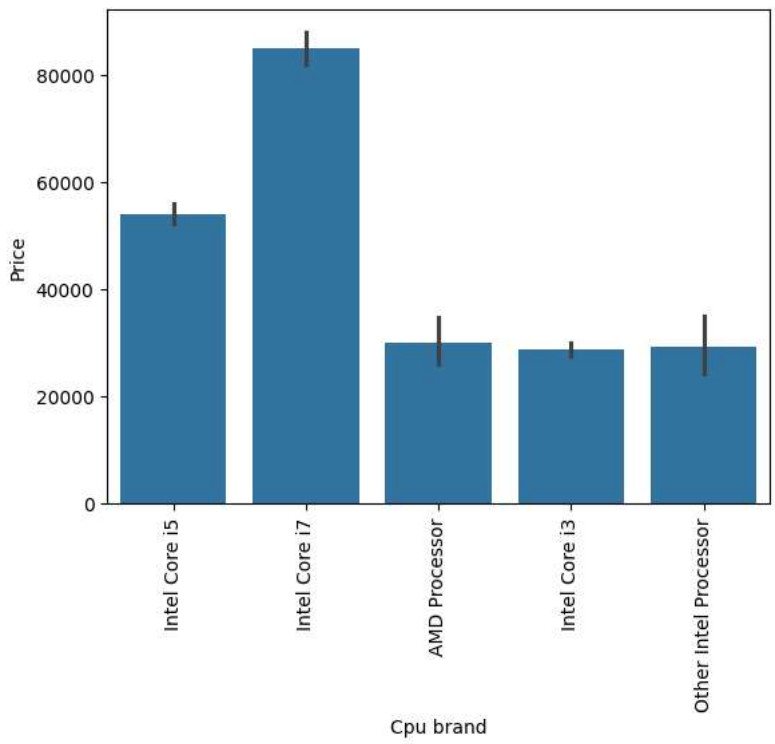
Next steps: Generate code with df | View recommended plots | New interactive sheet

```
df['Cpu brand'].value_counts().plot(kind='bar')
```

```
<Axes: xlabel='Cpu brand'>
```




```
sns.barplot(x=df['Cpu brand'],y=df['Price'])
plt.xticks(rotation='vertical')
plt.show()
```




```
df.drop(columns=['Cpu','Cpu Name'],inplace=True)
```

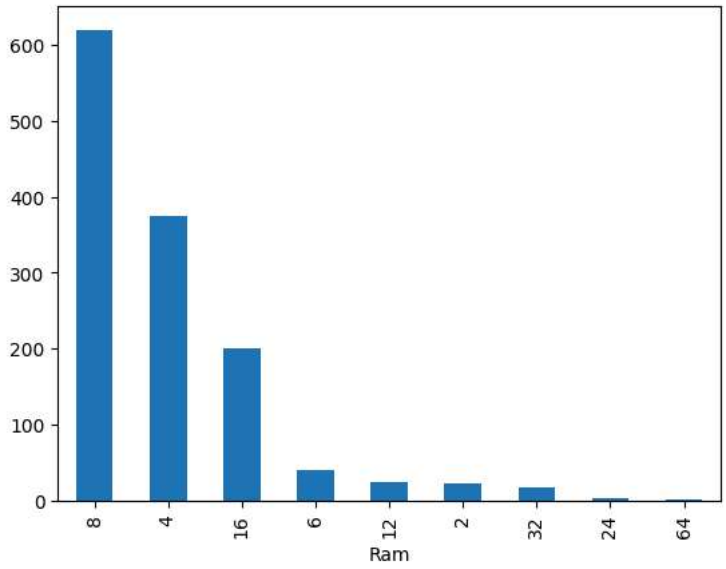
```
df.head()
```

	Company	TypeName	Ram	Memory	Gpu	OpSys	Weight	Price	Touchscreen	Ips	ppi	Cpu brand
0	Apple	Ultrabook	8	128GB SSD	Intel Iris Plus Graphics 640	macOS	1.37	71378.6832	0	1	226.983005	Intel Core i5
1	Apple	Ultrabook	8	128GB Flash Storage	Intel HD Graphics 6000	macOS	1.34	47895.5232	0	0	127.677940	Intel Core i5
2	HP	Notebook	8	256GB SSD	Intel HD Graphics 620	No OS	1.86	30636.0000	0	0	141.211998	Intel Core i5

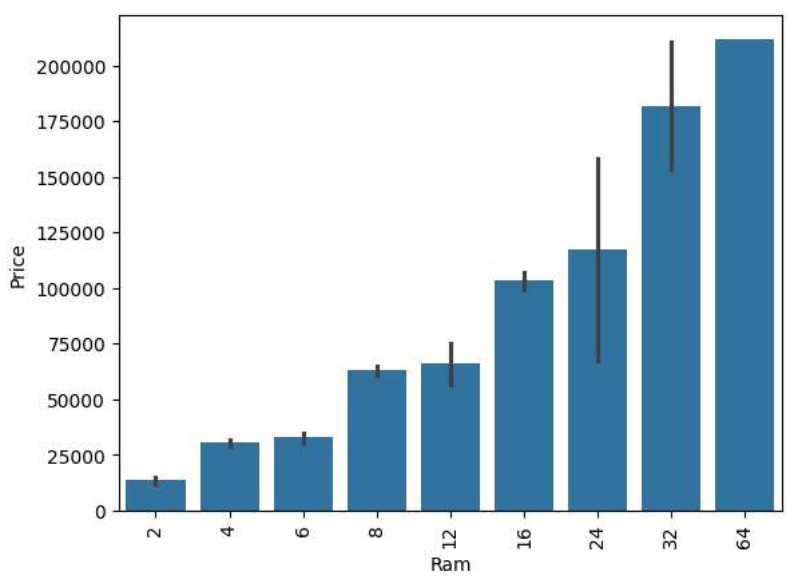
Next steps: Generate code with df | View recommended plots | New interactive sheet

```
df['Ram'].value_counts().plot(kind='bar')
```

```
<Axes: xlabel='Ram'>
```




```
sns.barplot(x=df['Ram'],y=df['Price'])
plt.xticks(rotation='vertical')
plt.show()
```




```
df['Memory'].value_counts()
```

Memory	count
256GB SSD	412
1TB HDD	223
500GB HDD	132
512GB SSD	118
128GB SSD + 1TB HDD	94
128GB SSD	76
256GB SSD + 1TB HDD	73
32GB Flash Storage	38
2TB HDD	16
64GB Flash Storage	15
512GB SSD + 1TB HDD	14
1TB SSD	14
256GB SSD + 2TB HDD	10
1.0TB Hybrid	9
256GB Flash Storage	8
16GB Flash Storage	7
32GB SSD	6
180GB SSD	5
128GB Flash Storage	4
512GB SSD + 2TB HDD	3
16GB SSD	3
512GB Flash Storage	2
1TB SSD + 1TB HDD	2
256GB SSD + 500GB HDD	2
128GB SSD + 2TB HDD	2
256GB SSD + 256GB SSD	2
512GB SSD + 256GB SSD	1
512GB SSD + 512GB SSD	1
64GB Flash Storage + 1TB HDD	1
1TB HDD + 1TB HDD	1
32GB HDD	1
64GB SSD	1
128GB HDD	1
240GB SSD	1
8GB SSD	1
508GB Hybrid	1
1.0TB HDD	1
512GB SSD + 1.0TB Hybrid	1
256GB SSD + 1.0TB Hybrid	1

dtype: int64

```
df['Memory'] = df['Memory'].astype(str).replace('\.0', '', regex=True)
df["Memory"] = df["Memory"].str.replace('GB', '')
df["Memory"] = df["Memory"].str.replace('TB', '000')
new = df["Memory"].str.split("+", n = 1, expand = True)

df["first"]= new[0]
```

```
df["first"]=df["first"].str.strip()

df["second"]= new[1]

df["Layer1HDD"] = df["first"].apply(lambda x: 1 if "HDD" in x else 0)
df["Layer1SSD"] = df["first"].apply(lambda x: 1 if "SSD" in x else 0)
df["Layer1Hybrid"] = df["first"].apply(lambda x: 1 if "Hybrid" in x else 0)
df["Layer1Flash_Storage"] = df["first"].apply(lambda x: 1 if "Flash Storage" in x else 0)

# Modified this line to extract only digits
df['first'] = df['first'].str.findall(r'\d+').str[0]

df["second"].fillna("0", inplace = True)

df["Layer2HDD"] = df["second"].apply(lambda x: 1 if "HDD" in x else 0)
df["Layer2SSD"] = df["second"].apply(lambda x: 1 if "SSD" in x else 0)
df["Layer2Hybrid"] = df["second"].apply(lambda x: 1 if "Hybrid" in x else 0)
df["Layer2Flash_Storage"] = df["second"].apply(lambda x: 1 if "Flash Storage" in x else 0)

# Modified this line to extract only digits
df['second'] = df['second'].str.findall(r'\d+').str[0].fillna('0')


df["first"] = df["first"].astype(int)
df["second"] = df["second"].astype(int)

df["HDD"]=(df["first"]*df["Layer1HDD"]+df["second"]*df["Layer2HDD"])
df["SSD"]=(df["first"]*df["Layer1SSD"]+df["second"]*df["Layer2SSD"])
df["Hybrid"]=(df["first"]*df["Layer1Hybrid"]+df["second"]*df["Layer2Hybrid"])
df["Flash_Storage"]=(df["first"]*df["Layer1Flash_Storage"]+df["second"]*df["Layer2Flash_Storage"])

df.drop(columns=['first', 'second', 'Layer1HDD', 'Layer1SSD', 'Layer1Hybrid',
       'Layer1Flash_Storage', 'Layer2HDD', 'Layer2SSD', 'Layer2Hybrid',
       'Layer2Flash_Storage'],inplace=True)
```

```
<>:1: SyntaxWarning: invalid escape sequence '\.'
<>:1: SyntaxWarning: invalid escape sequence '\.'
/tmp/ipython-input-33559402.py:1: SyntaxWarning: invalid escape sequence '\.'
  df['Memory'] = df['Memory'].astype(str).replace('\.0', '', regex=True)
/tmp/ipython-input-33559402.py:19: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignm
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting valu

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].me


  df["second"].fillna("0", inplace = True)
```

```
df.sample(5)
```

	Company	TypeName	Ram	Memory	Gpu	OpSys	Weight	Price	Touchscreen	Ips	ppi	Cpu brand	HDD	SSD	Hybrid	Flash_S
105	HP	Notebook	6	256 SSD	Nvidia GeForce 940MX	Windows 10	1.58	35111.520	0	1	157.350512	Intel Core i5	0	256	0	
520	Lenovo	Gaming	8	256 SSD + 1000 HDD	Nvidia GeForce GTX 1060	Windows 10	3.20	74538.720	0	1	141.211998	Intel Core i7	1000	256	0	

```
df.drop(columns=['Memory'],inplace=True)
```

```
df.head()
```

	Company	TypeName	Ram	Gpu	OpSys	Weight	Price	Touchscreen	Ips	ppi	Cpu brand	HDD	SSD	Hybrid	Flash_Storage
0	Apple	Ultrabook	8	Intel Iris Plus Graphics 640	macOS	1.37	71378.6832	0	1	226.983005	Intel Core i5	0	128	0	0
1	Apple	Ultrabook	8	Intel HD Graphics 6000	macOS	1.34	47895.5232	0	0	127.677940	Intel Core i5	0	0	0	128

Next steps: Generate code with df | View recommended plots | New interactive sheet

```
df.corr(numeric_only=True)['Price']
```

	Price
Ram	0.743007
Weight	0.210370
Price	1.000000
Touchscreen	0.191226
Ips	0.252208
ppi	0.473487
HDD	-0.096441
SSD	0.670799
Hybrid	0.007989
Flash_Storage	-0.040511

dtype: float64

```
df.drop(columns=['Hybrid','Flash_Storage'],inplace=True)
```

```
df.head()
```

	Company	TypeName	Ram	Gpu	OpSys	Weight	Price	Touchscreen	Ips	ppi	Cpu brand	HDD	SSD
0	Apple	Ultrabook	8	Intel Iris Plus Graphics 640	macOS	1.37	71378.6832	0	1	226.983005	Intel Core i5	0	128
1	Apple	Ultrabook	8	Intel HD Graphics 6000	macOS	1.34	47895.5232	0	0	127.677940	Intel Core i5	0	0
2	HP	Notebook	8	Intel HD Graphics 620	No OS	1.86	30636.0000	0	0	141.211998	Intel Core i5	0	256
3	Apple	Ultrabook	16	AMD Radeon Pro 455	macOS	1.83	135195.3360	0	1	220.534624	Intel Core i7	0	512
4	Apple	Ultrabook	8	Intel Iris Plus Graphics 650	macOS	1.37	96095.8080	0	1	226.983005	Intel Core i5	0	256

Next steps: Generate code with df | View recommended plots | New interactive sheet

```
df['Gpu'].value_counts()
```

Gpu	count
Intel HD Graphics 620	281
Intel HD Graphics 520	185
Intel UHD Graphics 620	68
Nvidia GeForce GTX 1050	66
Nvidia GeForce GTX 1060	48
...	...
AMD Radeon R5 520	1
AMD Radeon R7	1
Intel HD Graphics 540	1
AMD Radeon 540	1
ARM Mali T860 MP4	1

110 rows × 1 columns

dtype: int64

```
df['Gpu brand'] = df['Gpu'].apply(lambda x:x.split()[0])
```

```
df.head()
```

	Company	TypeName	Ram	Gpu	OpSys	Weight	Price	Touchscreen	Ips	ppi	Cpu brand	HDD	SSD	Gpu brand
0	Apple	Ultrabook	8	Intel Iris Plus Graphics 640	macOS	1.37	71378.6832	0	1	226.983005	Intel Core i5	0	128	Intel
1	Apple	Ultrabook	8	Intel HD Graphics 6000	macOS	1.34	47895.5232	0	0	127.677940	Intel Core i5	0	0	Intel
2	HP	Notebook	8	Intel HD Graphics 620	No OS	1.86	30636.0000	0	0	141.211998	Intel Core i5	0	256	Intel

Next steps: Generate code with df | View recommended plots | New interactive sheet

```
df['Gpu brand'].value_counts()
```

Gpu brand	count
Intel	722
Nvidia	400
AMD	180
ARM	1

dtype: int64

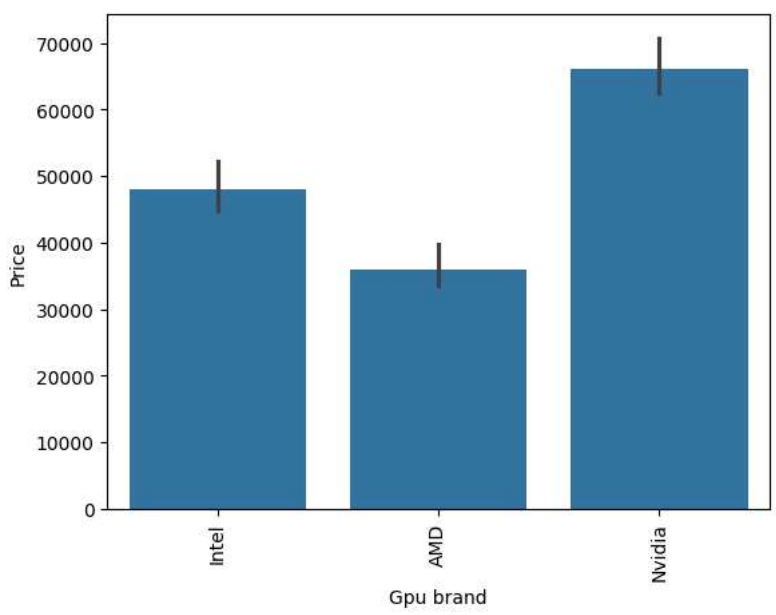
```
df = df[df['Gpu brand'] != 'ARM']
```

```
df['Gpu brand'].value_counts()
```

Gpu brand	count
Intel	722
Nvidia	400
AMD	180

dtype: int64

```
sns.barplot(x=df['Gpu brand'],y=df['Price'],estimator=np.median)
plt.xticks(rotation='vertical')
plt.show()
```



```
df.drop(columns=['Gpu'],inplace=True)
```

```
df.head()
```

	Company	TypeName	Ram	OpSys	Weight	Price	Touchscreen	Ips	ppi	Cpu brand	HDD	SSD	Gpu brand
0	Apple	Ultrabook	8	macOS	1.37	71378.6832	0	1	226.983005	Intel Core i5	0	128	Intel
1	Apple	Ultrabook	8	macOS	1.34	47895.5232	0	0	127.677940	Intel Core i5	0	0	Intel
2	HP	Notebook	8	No OS	1.86	30636.0000	0	0	141.211998	Intel Core i5	0	256	Intel
3	Apple	Ultrabook	16	macOS	1.83	135195.3360	0	1	220.534624	Intel Core i7	0	512	AMD
4	Apple	Ultrabook	8	macOS	1.37	96095.8080	0	1	226.983005	Intel Core i5	0	256	Intel

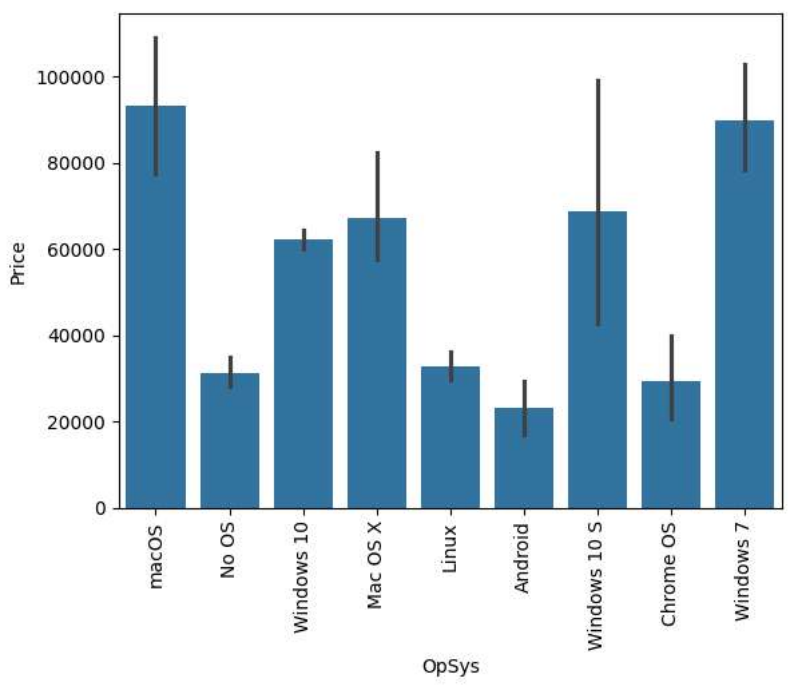
Next steps: Generate code with df | View recommended plots | New interactive sheet

```
df['OpSys'].value_counts()
```

OpSys	count
Windows 10	1072
No OS	66
Linux	62
Windows 7	45
Chrome OS	26
macOS	13
Mac OS X	8
Windows 10 S	8
Android	2

dtype: int64

```
sns.barplot(x=df['OpSys'],y=df['Price'])
plt.xticks(rotation='vertical')
plt.show()
```



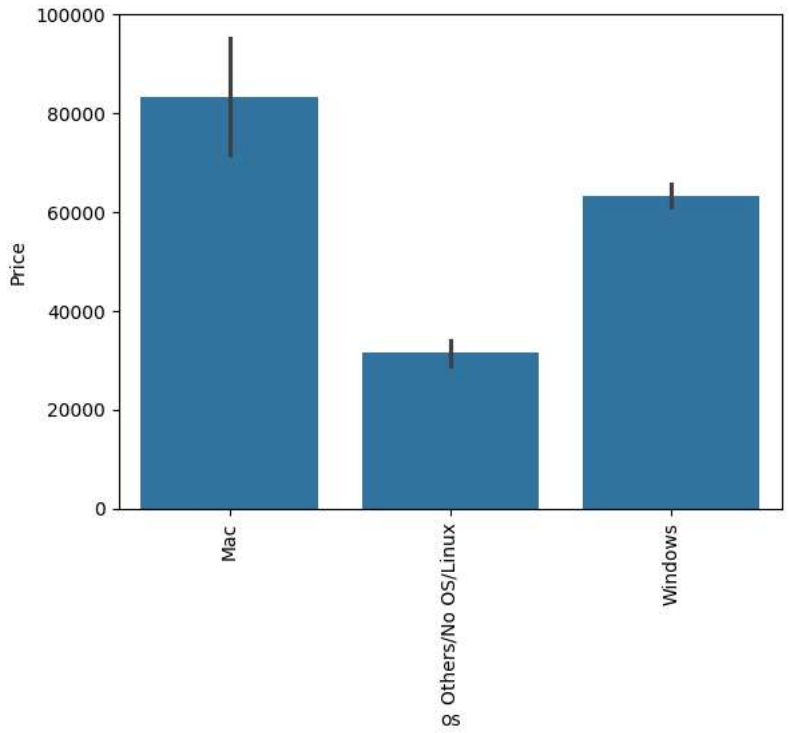

```
def cat_os(inp):
    if inp == 'Windows 10' or inp == 'Windows 7' or inp == 'Windows 10 S':
```

```
        return 'Windows'
    elif inp == 'macOS' or inp == 'Mac OS X':
        return 'Mac'
    else:
        return 'Others/No OS/Linux'

df['os'] = df['OpSys'].apply(cat_os)

df.head()
```

	Company	TypeName	Ram	OpSys	Weight	Price	Touchscreen	Ips	ppi	Cpu brand	HDD	SSD	Gpu brand	os
0	Apple	Ultrabook	8	macOS	1.37	71378.6832	0	1	226.983005	Intel Core i5	0	128	Intel	Mac
1	Apple	Ultrabook	8	macOS	1.34	47895.5232	0	0	127.677940	Intel Core i5	0	0	Intel	Mac
2	HP	Notebook	8	No OS	1.86	30636.0000	0	0	141.211998	Intel Core i5	0	256	Intel	Others/No OS/Linux

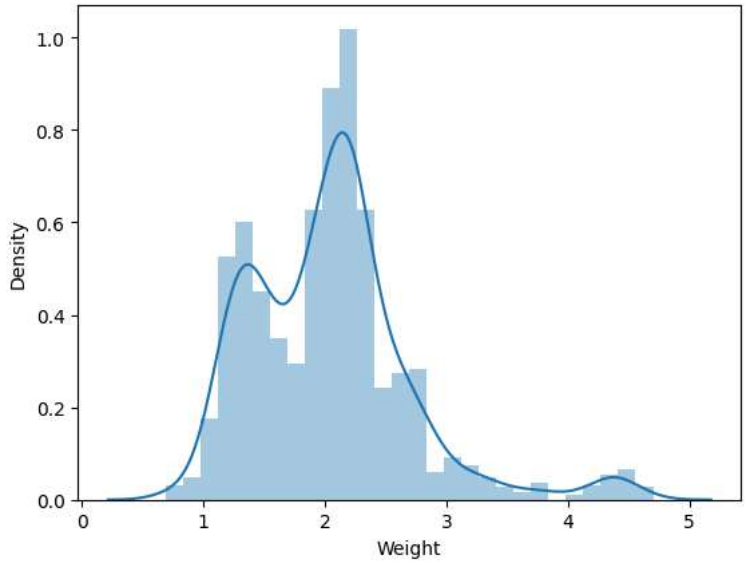
Next steps: Generate code with df | View recommended plots | New interactive sheet

```
df.drop(columns=['OpSys'],inplace=True)

sns.barplot(x=df['os'],y=df['Price'])
plt.xticks(rotation='vertical')
plt.show()
```



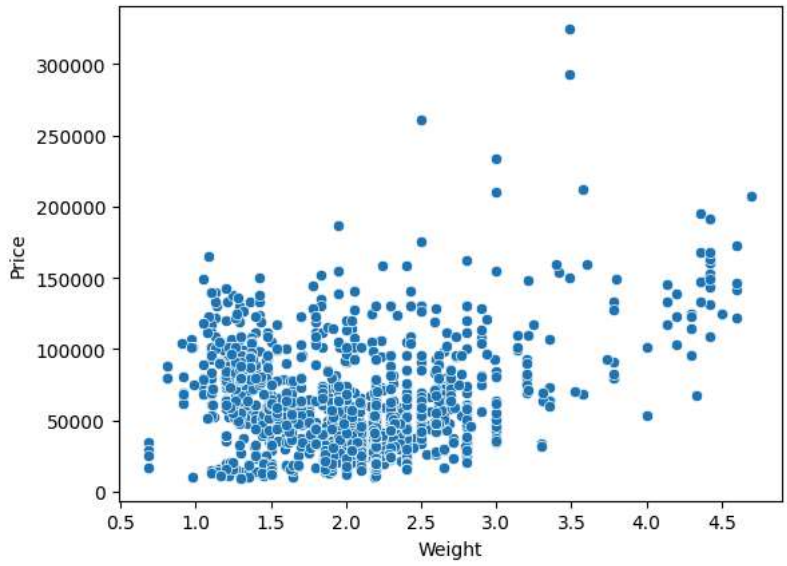

```
sns.distplot(df['Weight'])
```

```
/tmp/ipython-input-1125578356.py:1: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with
similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

  sns.distplot(df['Weight'])
<Axes: xlabel='Weight', ylabel='Density'>
```




```
sns.scatterplot(x=df['Weight'],y=df['Price'])
```

```
<Axes: xlabel='Weight', ylabel='Price'>
```



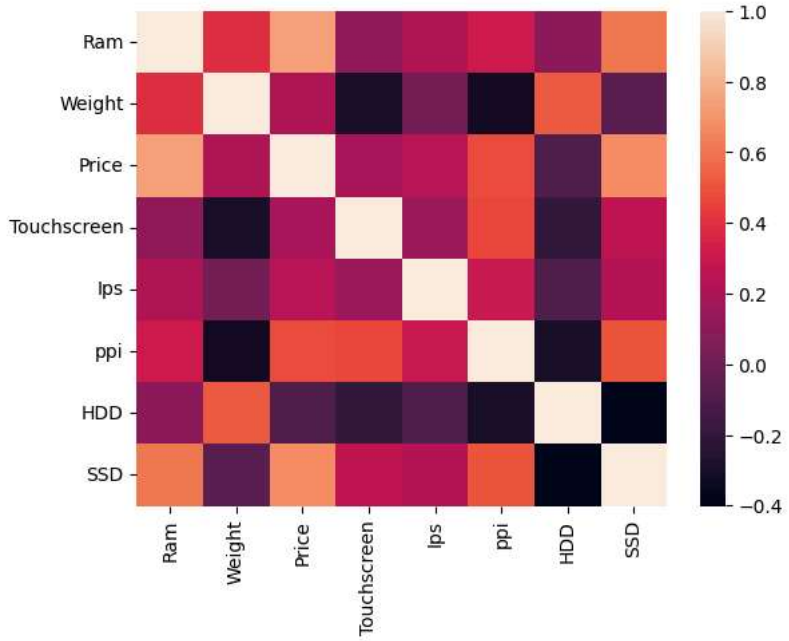

```
df.corr(numeric_only=True)['Price']
```

	Price
Ram	0.742905
Weight	0.209867
Price	1.000000
Touchscreen	0.192917
Ips	0.253320
ppi	0.475368
HDD	-0.096891
SSD	0.670660

dtype: float64

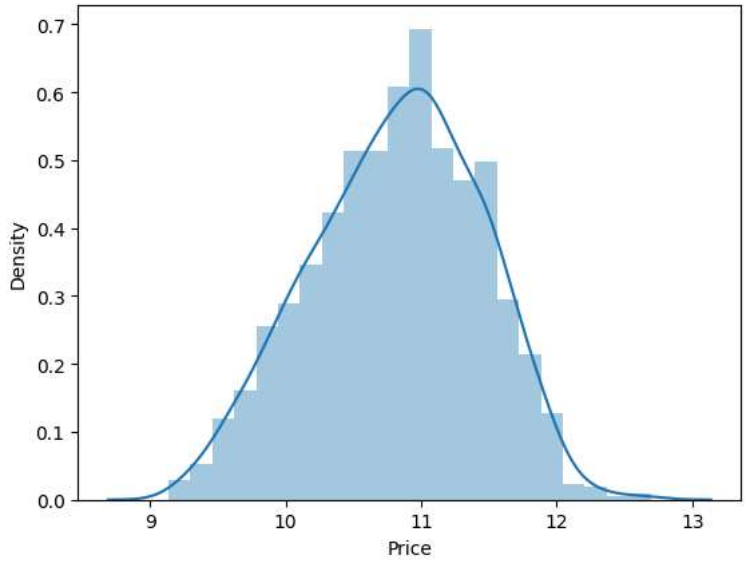
```
sns.heatmap(df.corr(numeric_only=True))
```

```
<Axes: >
```




```
sns.distplot(np.log(df['Price']))
```

```
/tmp/ipython-input-3556049916.py:1: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with
similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

  sns.distplot(np.log(df['Price']))
<Axes: xlabel='Price', ylabel='Density'>
```




```
X = df.drop(columns=['Price'])
y = np.log(df['Price'])
```

X

	Company	TypeName	Ram	Weight	Touchscreen	Ips	ppi	Cpu brand	HDD	SSD	Gpu brand	os
0	Apple	Ultrabook	8	1.37	0	1	226.983005	Intel Core i5	0	128	Intel	Mac
1	Apple	Ultrabook	8	1.34	0	0	127.677940	Intel Core i5	0	0	Intel	Mac
2	HP	Notebook	8	1.86	0	0	141.211998	Intel Core i5	0	256	Intel	Others/No OS/Linux
3	Apple	Ultrabook	16	1.83	0	1	220.534624	Intel Core i7	0	512	AMD	Mac
4	Apple	Ultrabook	8	1.37	0	1	226.983005	Intel Core i5	0	256	Intel	Mac
...	...	...	...	...	...	...	...	...	...	...	...	...
1298	Lenovo	2 in 1 Convertible	4	1.80	1	1	157.350512	Intel Core i7	0	128	Intel	Windows
1299	Lenovo	2 in 1 Convertible	16	1.30	1	1	276.053530	Intel Core i7	0	512	Intel	Windows
1300	Lenovo	Notebook	2	1.50	0	0	111.935204	Other Intel Processor	0	0	Intel	Windows

Next steps: Generate code with X | View recommended plots | New interactive sheet

y

	Price
0	11.175755
1	10.776777
2	10.329931
3	11.814476
4	11.473101
...	...
1298	10.433899
1299	11.288115
1300	9.409283
1301	10.614129
1302	9.886358

1302 rows × 1 columns

dtype: float64

```
from sklearn.model_selection import train_test_split
X_train,X_test,y_train,y_test = train_test_split(X,y,test_size=0.15,random_state=2)
```

```
X_train
```

	Company	TypeName	Ram	Weight	Touchscreen	Ips	ppi	Cpu brand	HDD	SSD	Gpu brand	os
183	Toshiba	Notebook	8	2.00	0	0	100.454670	Intel Core i5	0	128	Intel	Windows
1141	MSI	Gaming	8	2.40	0	0	141.211998	Intel Core i7	1000	128	Nvidia	Windows
1049	Asus	Netbook	4	1.20	0	0	135.094211	Other Intel Processor	0	0	Intel	Others/No OS/Linux
1020	Dell	2 in 1 Convertible	4	2.08	1	1	141.211998	Intel Core i3	1000	0	Intel	Windows
878	Dell	Notebook	4	2.18	0	0	141.211998	Intel Core i5	1000	128	Nvidia	Windows
...	...	...	...	...	...	...	...	...	...	...	...	...
466	Acer	Notebook	4	2.20	0	0	100.454670	Intel Core i3	500	0	Nvidia	Windows
299	Asus	Ultrabook	16	1.63	0	0	141.211998	Intel Core i7	0	512	Nvidia	Windows
493	Acer	Notebook	8	2.20	0	0	100.454670	AMD Processor	1000	0	AMD	Windows
527	Lenovo	Notebook	8	2.20	0	0	100.454670	Intel Core i3	2000	0	Nvidia	Others/No OS/Linux

Next steps: Generate code with X_train | View recommended plots | New interactive sheet

```
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import OneHotEncoder
from sklearn.metrics import r2_score,mean_absolute_error


from sklearn.linear_model import LinearRegression,Ridge,Lasso
from sklearn.neighbors import KNeighborsRegressor
from sklearn.tree import DecisionTreeRegressor
from sklearn.ensemble import RandomForestRegressor,GradientBoostingRegressor,AdaBoostRegressor,ExtraTreesRegressor
from sklearn.svm import SVR
from xgboost import XGBRegressor
```

Linear regression

```
step1 = ColumnTransformer(transformers=[
    ('col_tnf',OneHotEncoder(drop='first'),[0,1,7,10,11])
],remainder='passthrough')
```

```
step2 = LinearRegression()

pipe = Pipeline([
    ('step1',step1),
    ('step2',step2)
])

pipe.fit(X_train,y_train)

y_pred = pipe.predict(X_test)

print('R2 score',r2_score(y_test,y_pred))
print('MAE',mean_absolute_error(y_test,y_pred))
```

```
R2 score 0.807327744841864
MAE 0.21017827976428802
```

Ridge Regression

```
step1 = ColumnTransformer(transformers=[
    ('col_tnf',OneHotEncoder(drop='first'),[0,1,7,10,11])
],remainder='passthrough')

step2 = Ridge(alpha=10)

pipe = Pipeline([
    ('step1',step1),
    ('step2',step2)
])

pipe.fit(X_train,y_train)

y_pred = pipe.predict(X_test)

print('R2 score',r2_score(y_test,y_pred))
print('MAE',mean_absolute_error(y_test,y_pred))
```

```
R2 score 0.8127331031311809
MAE 0.20926802242582968
```

Lasso Regression

```
step1 = ColumnTransformer(transformers=[
    ('col_tnf',OneHotEncoder(drop='first'),[0,1,7,10,11])
],remainder='passthrough')

step2 = Lasso(alpha=0.001)

pipe = Pipeline([
    ('step1',step1),
    ('step2',step2)
])

pipe.fit(X_train,y_train)

y_pred = pipe.predict(X_test)

print('R2 score',r2_score(y_test,y_pred))
print('MAE',mean_absolute_error(y_test,y_pred))
```

```
R2 score 0.8071853945317105
MAE 0.21114361613472565
```

KNN

```
step1 = ColumnTransformer(transformers=[
    ('col_tnf',OneHotEncoder(drop='first'),[0,1,7,10,11])
],remainder='passthrough')

step2 = KNeighborsRegressor(n_neighbors=3)
```

```
pipe = Pipeline([
    ('step1',step1),
    ('step2',step2)
])

pipe.fit(X_train,y_train)

y_pred = pipe.predict(X_test)

print('R2 score',r2_score(y_test,y_pred))
print('MAE',mean_absolute_error(y_test,y_pred))
```

```
R2 score 0.8017673664034364
MAE 0.19346118183798544
```

Decision Tree

```
step1 = ColumnTransformer(transformers=[
    ('col_tnf',OneHotEncoder(drop='first'),[0,1,7,10,11])
],remainder='passthrough')

step2 = DecisionTreeRegressor(max_depth=8)

pipe = Pipeline([
    ('step1',step1),
    ('step2',step2)
])

pipe.fit(X_train,y_train)

y_pred = pipe.predict(X_test)

print('R2 score',r2_score(y_test,y_pred))
print('MAE',mean_absolute_error(y_test,y_pred))
```

```
R2 score 0.8457037672678894
MAE 0.18010727725713122
```

SVM

```
step1 = ColumnTransformer(transformers=[
    ('col_tnf',OneHotEncoder(drop='first'),[0,1,7,10,11])
],remainder='passthrough')

step2 = SVR(kernel='rbf',C=10000,epsilon=0.1)

pipe = Pipeline([
    ('step1',step1),
    ('step2',step2)
])

pipe.fit(X_train,y_train)

y_pred = pipe.predict(X_test)

print('R2 score',r2_score(y_test,y_pred))
print('MAE',mean_absolute_error(y_test,y_pred))
```

```
R2 score 0.8083180902272435
MAE 0.20239059427315706
```

Random Forest

```
step1 = ColumnTransformer(transformers=[
    ('col_tnf',OneHotEncoder(drop='first'),[0,1,7,10,11]
],remainder='passthrough')

step2 = RandomForestRegressor(n_estimators=100,
                              random_state=3,
                              max_samples=0.5,
                              max_features=0.75,
```

```
                              max_depth=15)

pipe = Pipeline([
    ('step1',step1),
    ('step2',step2)
])

pipe.fit(X_train,y_train)

y_pred = pipe.predict(X_test)

print('R2 score',r2_score(y_test,y_pred))
```

```
R2 score 0.8873402378382488
MAE 0.15860130110457718
```

ExtraTrees

```
step1 = ColumnTransformer(transformers=[
    ('col_tnf',OneHotEncoder(drop='first'),[0,1,7,10,11])
],remainder='passthrough')

step2 = ExtraTreesRegressor(n_estimators=100,
                            random_state=3,
                            max_samples=0.5,
                            max_features=0.75,
                            max_depth=15,
                            bootstrap=True)

pipe = Pipeline([
    ('step1',step1),
    ('step2',step2)
])

pipe.fit(X_train,y_train)

y_pred = pipe.predict(X_test)

print('R2 score',r2_score(y_test,y_pred))
print('MAE',mean_absolute_error(y_test,y_pred))
```

```
R2 score 0.8850720167552375
MAE 0.16154538000217084
```

AdaBoost

```
step1 = ColumnTransformer(transformers=[
    ('col_tnf',OneHotEncoder(drop='first'),[0,1,7,10,11])
],remainder='passthrough')

step2 = AdaBoostRegressor(n_estimators=15,learning_rate=1.0)

pipe = Pipeline([
    ('step1',step1),
    ('step2',step2)
])

pipe.fit(X_train,y_train)

y_pred = pipe.predict(X_test)

print('R2 score',r2_score(y_test,y_pred))
print('MAE',mean_absolute_error(y_test,y_pred))
```

```
R2 score 0.7854544057327982
```